

[← Back to course](#)

Exam 2

- The exam time is from 12:45 - 2:00. The time on Crowdmark is different for students with testing accommodations.
- Images must be uploaded as JPG, PNG, or PDF. Images must be in the correct orientation (i.e., not rotated).
- There are 80 possible points but the exam will be out of 75.
- If your handwriting is not legible, please type your answers.



Submitted successfully!

Assignment submitted and ready for grading.

You can resubmit until **Thursday, March 19, 2026 3:00 pm (Eastern Daylight Time)** if needed.

[Edit and resubmit](#)

Q1 (8 points)

Submitted

Explain how the margin and kernel of an SVM work together to give us a linear model with nonlinear capabilities and automatic regularization.

In the dual SVM formulation, we use the kernel trick dependent on N Lagrange multipliers to build a relatively inexpensive non-linear representation of the linear vector dot-product of SVM inputs. Since we are trying to minimize our weights during the model fit, an

automatic "weight-decay" is observed due where the weights generally decrease as the model training progresses. These processes are built to maximize the margin "cushion" such that only support vectors lay on the edges.

Q2A (3 points)

Submitted

Given the objective for maximizing the margin:

$$\min_{b,w} \frac{1}{2} w^T w$$

subject to: $y_n(w^T x_n + b) = 1$

and given the standard form of the QP problem:

$$\min_{u \in \mathbf{R}^q} \frac{1}{2} u^T Q u + p^T u$$

subject to: $Au \geq c$

Explain what the component $\mathbf{u}$ represents and its usefulness to the linear hard-margin SVM algorithm.

$\mathbf{u} = [b, w_1, \dots, w_2]$, which accounts for both our bias and weights within the quadratic norm. This new form is useful because it is compatible with QP solvers such as CVXOPT. This solver computes the optimal weights in a convex minimization problem.

Q2B (3 points)

Submitted

Given the objective for maximizing the margin and standard QP problem in Q2A:

Explain what the component A represents and its usefulness to the linear hard-margin SVM algorithm.

A is the constraint matrix derived from $y_n(w^T x_n + b) = 1$. It allows us to map the primal equations constraint to the QP solver format (again CVXOPT).

Q2C (3 points)

Submitted

Given the objective for maximizing the margin and standard QP problem in Q2A:

Explain what the component $\mathbf{c}$ represents and its usefulness to the linear hard-margin SVM algorithm.

$\mathbf{c}$ is the -1 or $+1$ constraint along the margin surrounding the hyperplane line. Its useful for ensuring that the classifications lay on the margin or beyond the margin at all times.

Q3A (2 points)

Submitted

Given the soft-margin SVM objective:

$$\min_{b,w,\epsilon} \frac{1}{2} w^T w + C \sum_{n=1}^N \epsilon_n$$

subject to: $y_n(w^T x_n + b) \geq 1 - \epsilon_n, \epsilon_n \geq 0$

Explain what the component $\frac{1}{2} w^T w$ represents.

This is the quadratic norm (or primal equation). Its an objective version that is compatible with CVXOPT, an algorithm for finding optimal weights in a convex way.

Q3B (2 points)

Submitted

Given the soft-margin SVM objective:

$$\min_{b,w,\epsilon} \frac{1}{2} w^T w + C \sum_{n=1}^N \epsilon_n$$

subject to: $y_n(w^T x_n + b) \geq 1 - \epsilon_n, \epsilon_n \geq 0$

Explain what the component $C \sum_{n=1}^N \epsilon_n$ represents.

This is term that scales the tolerance for margin violations. Higher C parameter values get closer to hard-margin SVM constraints while low C values allow more points to cross to the wrong side of the margin for the sake of better generalizability.

Q4 (3 points)

Submitted

In the hard-margin SVM proofs, we have 2 versions of the satisfying condition.

Condition (1): $y_n(w^T x_n + b) = 1$ and **Condition (2):** $y_n(w^T x_n + b) \geq 1$.

Condition (1) represents ____; Condition (2) represents ____.

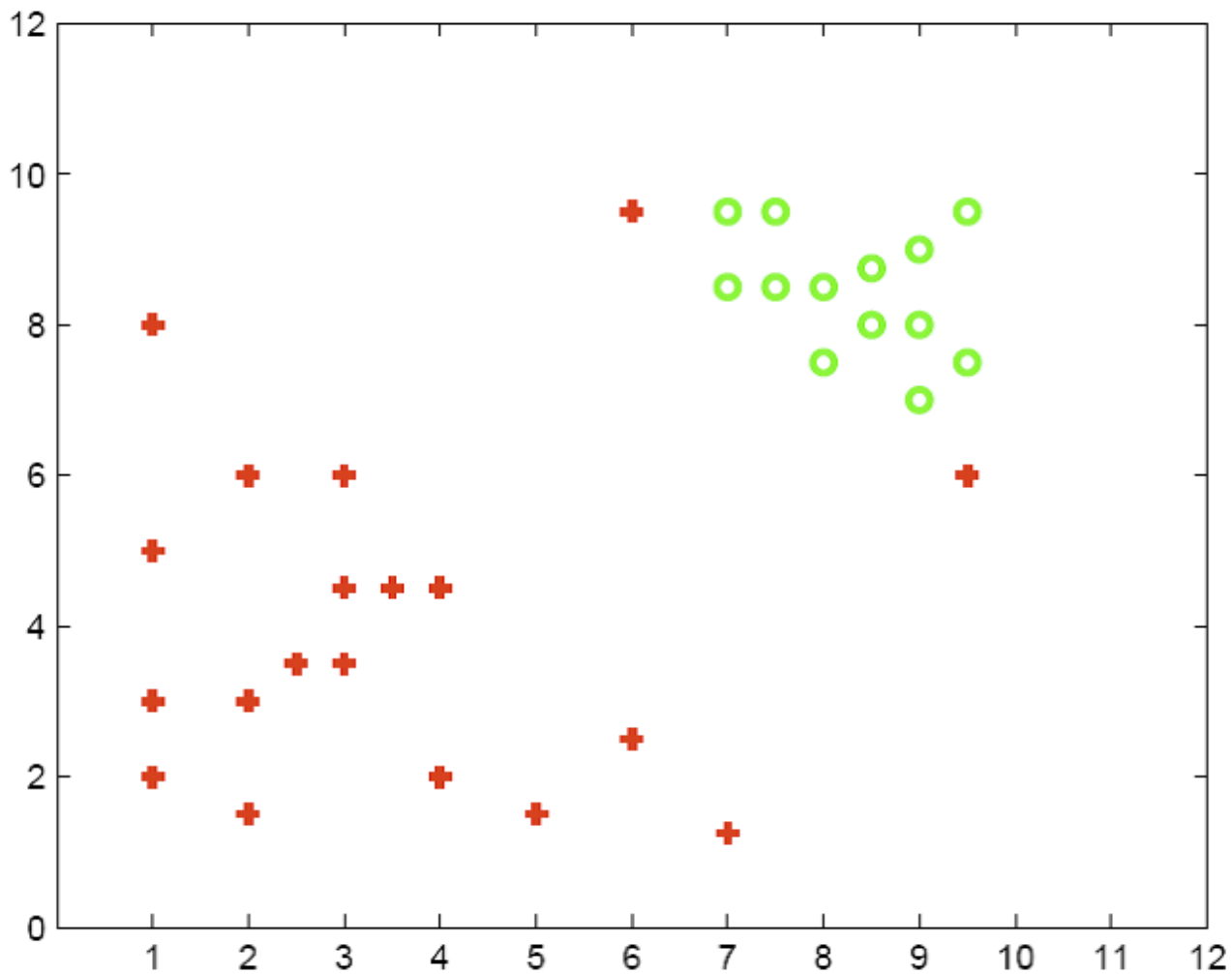
- ☒ data point x_n lies exactly on the margin boundary; every data point is correctly classified on or beyond the boundary
- ☐ data point x_n lies exactly on the positive class margin boundary; every data point is correctly classified beyond the boundary
- ☐ data point x_n lies exactly on the margin boundary; every data point is correctly classified beyond the boundary

Q5A (6 points)

Submitted

Using the figure below, where would the soft-margin SVM decision boundary be for very large values of C (i.e., $C \rightarrow \infty$)?

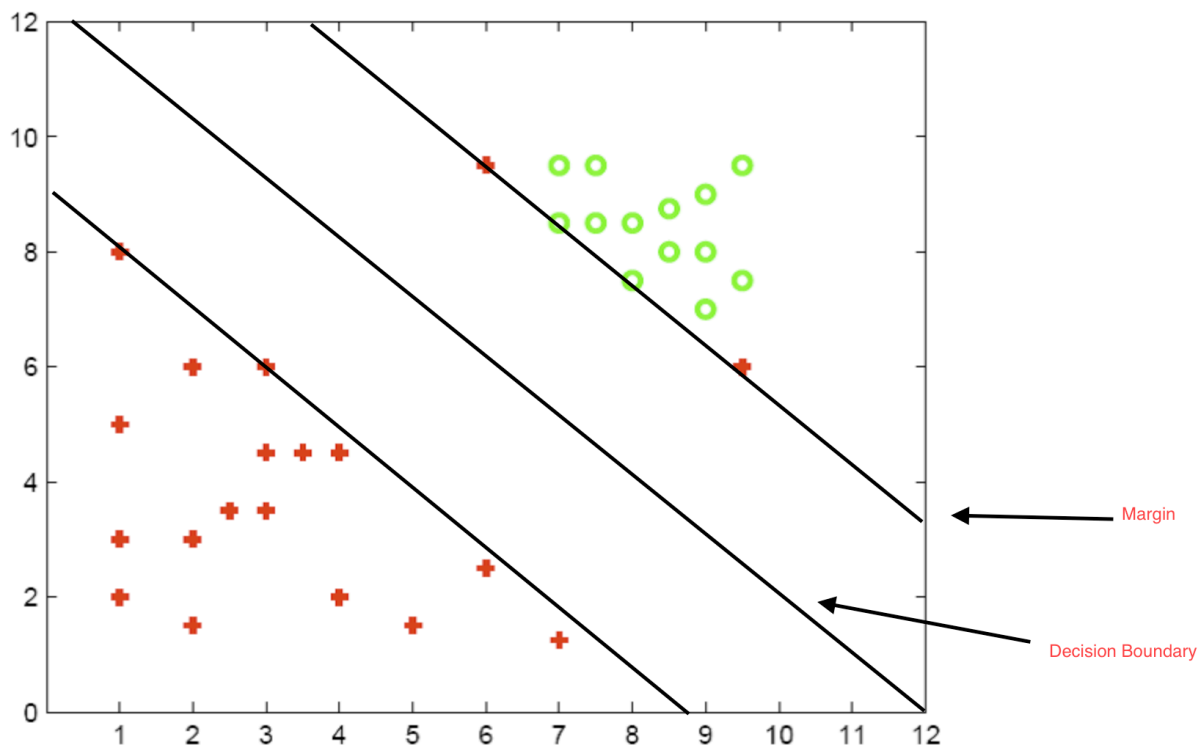
Draw the boundary (line and margin) on the figure. Justify your answer.



Q5A (6 points)

Using the figure below, where would the soft-margin SVM decision boundary be for very large values of C (i.e., $C \rightarrow \infty$)?

Draw the boundary (line and margin) on the figure. Justify your answer.

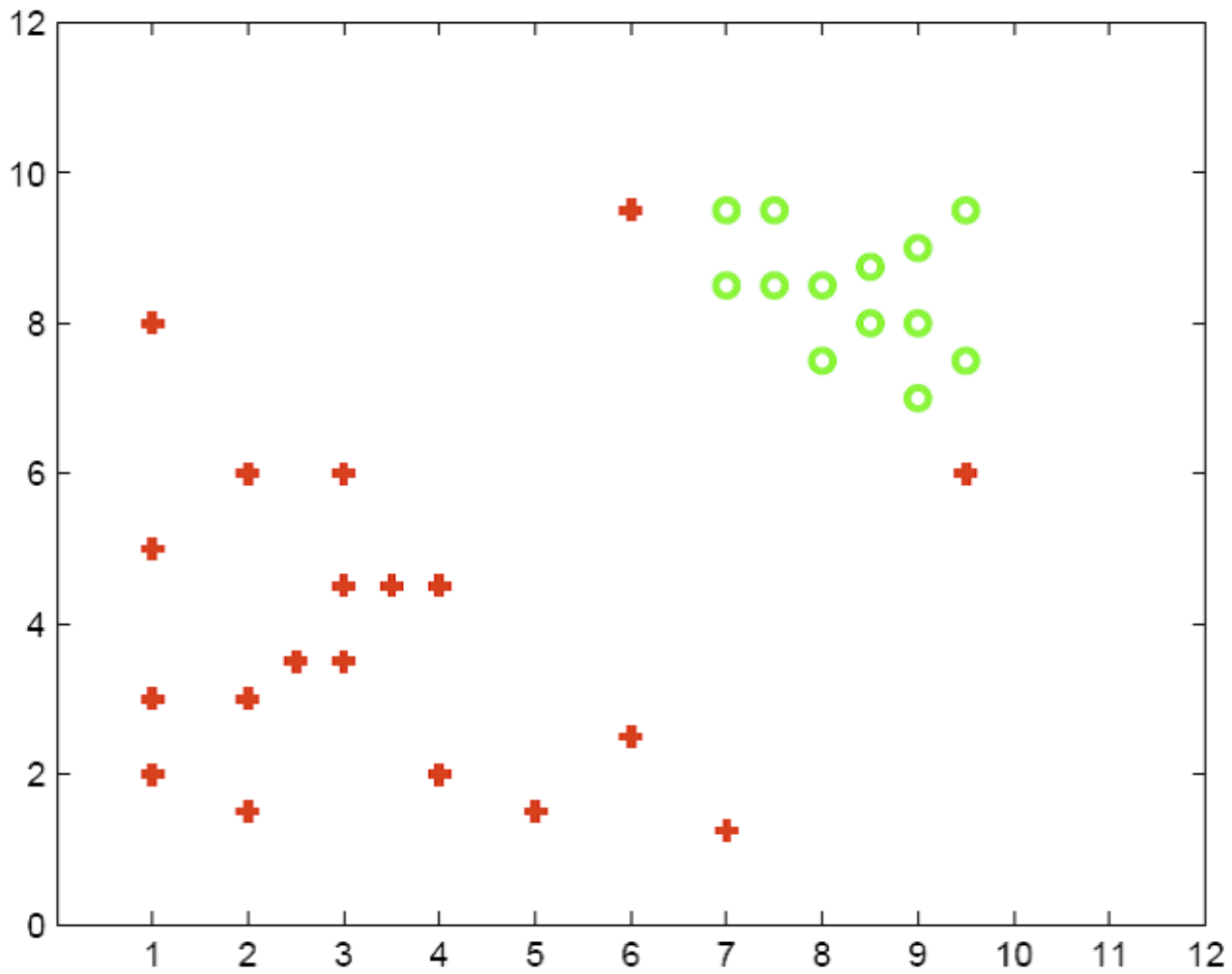


Since C is approaching infinity, this SVM is essentially hard-margin. Meaning that there are no margin violations present and support vectors lay exactly on the margin.

Q5B (6 points)

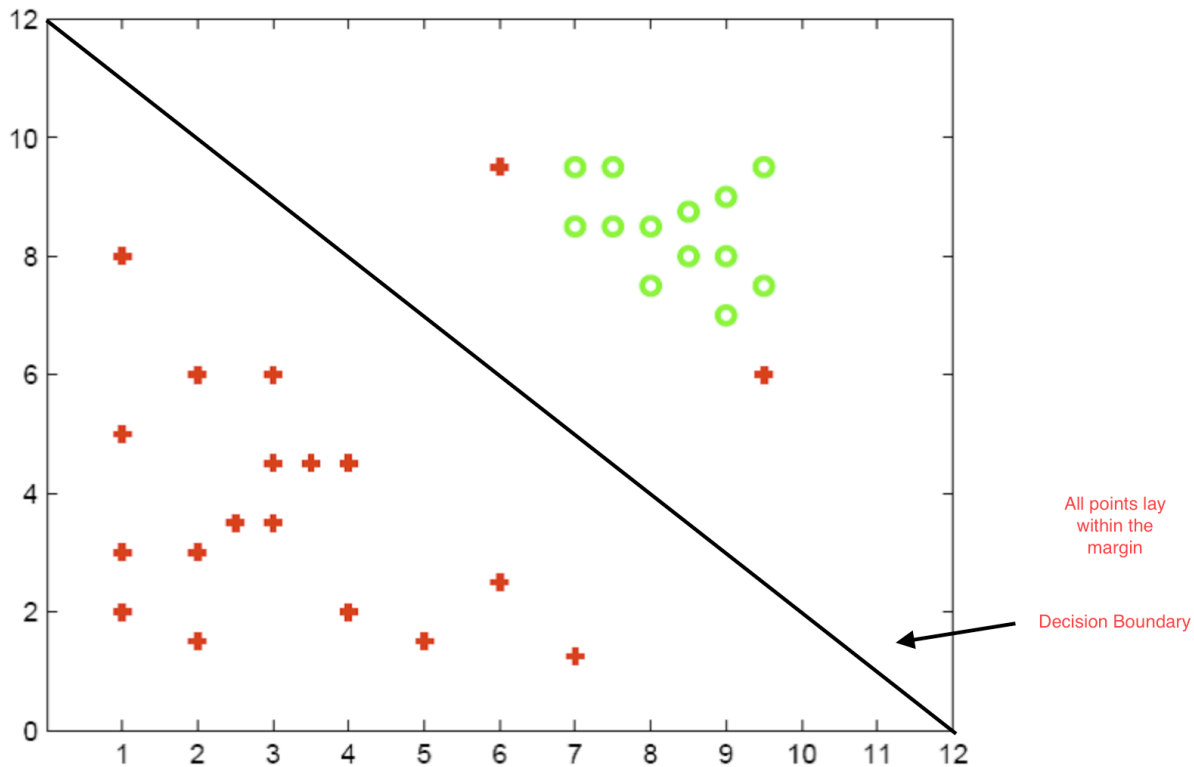
Submitted

For $C \approx 0$, draw where you would expect the decision boundary and margin to be. Justify your answer.



Q5B (6 points)

For $C \approx 0$, draw where you would expect the decision boundary and margin to be. Justify your answer.



Since $C = 0$, the margin is essentially the entire space of the data. That is, all points fall within the margin.

Q6A (2 points)

Submitted

For the number of samples N and the data dimension d , what are the number of variables we must optimize for in the **primal** hard-margin SVM formulation?

- ☐ d
- ☒ $d + 1$
- ☐ $d + 1 + N$
- ☐ N
- ☐ $N + 1$

Q6B (2 points)

Submitted

For the number of samples N and the data dimension d , what are the number of variables we must optimize for in the **dual** hard-margin SVM formulation?

- ☐ d
- ☐ $d + 1$
- ☐ $d + 1 + N$
- ☒ N
- ☐ $N + 1$

Q6C (6 points)

Submitted

In primal SVM, we optimize over the weight vector $\mathbf{w}$. In dual SVM, we optimize over the Lagrangian multipliers (α terms) of training vectors $(y_n y_m x_n x_m)$. Explain (A) why this shift is necessary and (B) how this allows us to apply the kernel trick. *Hint*: Q2A shows the primal objective which optimizes for $\mathbf{w}$. Recall from lecture the key difference between this primal version and the expanded Lagrangian equation of the dual version. What key piece of math was in that Lagrangian that allows us to apply different kernels?

(A) This shift is necessary because it allows us to implement the "kernel trick," which then allows us to compute non-linear vector dot-products without the expensive process of transforming all the inputs first.

Q7A (4 points)

Submitted

Why do we need to use the Error Backpropagation algorithm to compute the gradients needed for gradient descent during Neural Network training?

Because back propagation allows us to compute a loss function, which we can minimize to obtain the optimal NN weights. If we start on top of a convex curve, we gradually "roll" towards the optimal weight at the bottom of the curve.

Q7B (2 points)

Submitted

Using Error Backpropagation Through Time in RNNs causes *both* exploding gradients and vanishing gradients.

- ☐ True
- ☒ False

Q8A (3 points)

Submitted

For the following Neural Networks, list the key differences between the two architectures: Feedforward Neural Network (FFNN) and Recurrent Neural Network (RNN).

RNNs have a "recurrent link" that uses information from prior tokens in an input sequence to make predictions for the next token. This is a cyclical and expensive process in the hidden layer. FFNNs simply perform a one-time forward pass from the inputs to the hidden layer(s) to the output layer without looping in the middle.

Q8B (3 points)

Submitted

For the following Neural Networks, list the key differences between the two architectures: Recurrent Neural Network (RNN) and Long Short Term Memory (LSTM).

RNNs contain a recurrent link, which allows them to loop over and learn from the context of a sequence of tokens (e.g., words in a sentence). The fundamental flaw in this architecture is that the BPTT partial derivative multiplications caused old context to be "forgotten" as gradients approached 0. LSTM architecture solved this problem by implementing a context management layer and "gates" that filter which information to forget, input/add, and output.

Q8C (3 points)

Submitted

For the following Neural Networks, list the key differences between the two architectures: Recurrent Neural Network (RNN) and the basic Transformer.

RNNs have a recurrent link with learns over input sequences iteratively while transformer architecture parallelizes this process in an attention mechanism using mass vector dot products.

Q9A (2 points)

Submitted

Recall the formula for simple self-attention in a causal left-to-right transformer: $\text{softmax}(\text{mask}(\frac{QK^T}{\sqrt{d_k}}))V$.

I didn't mention this in class, but try to extrapolate from what you know about RNN and bidirectional RNN. What piece of the attention formula above can be removed to create a bidirectional transformer?



softmax



mask

**Q9B (6 points)**

Not answered

In your own words, list and briefly explain the 3 steps of simple self-attention (used in the simple self-attention layer of Transformers).

Q9C (6 points)

Not answered

Explain how Step 1 and Step 3 of the self-attention calculation (your answer for Q9B) are modified to reflect the idea of the *query*, *key*, and *value*.

Q10 (5 points)

Submitted

The Neural Network architectures can suffer from loss of initial information when processing longer input sequences. For (A) RNN and (B) Transformer, explain how these architectures can be modified to attempt to fix this information loss problem.

LSTM ameliorates this "vanishing gradient" problem by implementing logic gates and a context management layer that filters what context is forgotten, added, and outputted in the process.

[↑ Back to top](#)